

Annotated Bibliography

Ferguson, H. C., Greenfield, P., Axelrod, T., Baum, S., Conti, A., Crabtree, D., ... & White, R. L. (2010). *Astronomical data reduction and analysis for the next decade*. Space Telescope Science Institute (STScI). <https://www8.nationalacademies.org/astro2010/detailfiledisplay.aspx?id=426>

In this foundational white paper submitted to the Astronomy and Astrophysics Decadal Survey, the authors argue that the primary bottleneck in next-generation astronomy is no longer data collection, but rather data reduction and analytical throughput. They propose that current pipelines will structurally fail under the weight of incoming multi-petabyte datasets. It's noted that their software systems are falling behind the curve in the advancement of computer hardware - as the astronomical community often prioritized developing specific software to a new instrument or project-specific software in general. These siloed algorithms not only restrict access to a small group of researchers, but they also prevent the full utilization of available computational resources. This highlights a dire need to shift focus from delivering calibrated data to providing community-wide analytical tools. As an authoritative forecast from the Space Telescope Science Institute, this document establishes the historical context for the astronomical data deluge. For the symposium, this research mathematically justifies the necessary architectural leap from classical, distributed Big Data frameworks to quantum computing. While classical data processing currently relies on infrastructures like Hadoop, MapReduce, and PySpark to manage massive nightly datasets from observatories, this text demonstrates the absolute computational limits of those traditional nodes - specifically when tasked with complex 3-D modeling - thereby establishing the necessity of the quantum transition.

Garisto, D. (2025, January 15). Google measures quantum echoes on Willow quantum computer chip. *Scientific American*.

<https://www.scientificamerican.com/article/google-measures-quantum-echoes-on-willow-quantum-computer-chip/>

This journalistic article details Google's significant hardware breakthrough with the "Willow" chip, arguing that the measurement of "quantum echoes" drastically reduces error rates and increases viable quantum volume. Published in a highly reputable science magazine, the piece bridges the gap between theoretical physics and current technological capabilities. This source strengthens the symposium by providing immediate, real-world evidence of the "Quantum Escalation" and the recent 105-qubit milestone. Furthermore, it explicitly defines the physical hardware leap required to move beyond classical Big Data computing. Unlike classical servers running Hadoop in standard air-conditioned data centers, this article underscores that

these new architectures require superconducting circuits, Helium-3 cooling, and temperatures colder than deep space, emphasizing the extreme engineering required for the next generation of data science.

Google Gemini / Midjourney Generative Design Suite. (2026). Generative AI as a technical visualization tool for quantum architectural design.

<https://gemini.google.com/>

This entry represents the methodology used to generate the symposium's core visual assets, utilizing Large Language Models and generative design tools to translate abstract quantum mechanics into high-fidelity technical diagrams. Rather than serving as a source of primary research, these AI tools acted as a sophisticated graphic interface to visualize the internal logic of the 8-qubit circuit used in the Messier 51 galaxy reconstruction. Specifically, the tools were utilized to create the "Stuffing Probabilities" and "QML Execution" diagrams, which illustrate complex processes like Amplitude Embedding and Quantum Variational Layers that are otherwise invisible to the classical observer. For the symposium, this highlights a shift in technical communication, where AI-assisted imagery is used to bridge the gap between high-level Python code (PennyLane) and physical quantum hardware. This entry serves as a methodological disclosure, ensuring technical transparency by documenting that the presentation's visual frameworks were refined through AI synthesis to provide the audience with a concrete map of the project's quantum architecture.

Hawkins, M. (2024, September 6). U.S. will impose new export controls on quantum computing to stop China getting access to cutting-edge tech. *Fortune Asia*.

<https://fortune.com/asia/2024/09/05/us-to-impose-new-export-controls-china-quantum-computing-chips-semiconductors-ai/>

Hawkins reports on the immediate economic and political fallout of the 2024 export controls, arguing that the U.S. strategy has shifted from merely out-innovating competitors to actively restricting their access to next-generation quantum processing and cryogenic systems. As a contemporary business news piece, it offers a reliable perspective on global technology markets. This article proves that the geopolitical "Chip War" has officially expanded beyond classical silicon into the quantum sector. It reinforces the symposium's concept of geopolitical gatekeeping, demonstrating that access to QPU hardware is deliberately restricted by international policy. Consequently, the future of processing massive datasets relies not just on overcoming the software limits of MapReduce, but on securing strategic access to heavily guarded international hardware supply chains.

Heredge, J., Hill, C., Hollenberg, L., & Seviar, M. (2023). A quantum-enhanced support vector machine for galaxy classification. *RAS Techniques and Instruments*, 2(1), 752–762.

<https://doi.org/10.1093/rasti/rzad054>

In this peer-reviewed study, Heredge et al. argue that quantum kernels can effectively process highly complex, high-dimensional astronomical image data with competitive or superior accuracy compared to classical machine learning models. The researchers successfully mapped an analytical workload of 40,000 galaxies onto a Quantum Processing Unit (QPU) utilizing IBM's Qiskit. This paper serves as the empirical backbone for the symposium's technical demonstration. In a classical modern data architecture, a Support Vector Machine analyzing this volume of data would be distributed across multiple PySpark nodes. By providing a verified architectural blueprint for building quantum classification models, this research provides peer-reviewed proof for the presentation's claim that high-fidelity quantum hardware enables transformative efficiency for high-precision astrophysical tasks. While Heredge et al. utilize Qiskit, our symposium directly adapts these foundational concepts into a live environment using the PennyLane Quantum ML library. As demonstrated in the presentation's live code walkthrough, PennyLane acts as the crucial abstraction layer - using the `@qml.qnode` decorator - to translate classical Python code into quantum operations. Just as Heredge's team mapped galaxy data into quantum states, the symposium demonstration utilizes PennyLane's Amplitude Embedding to compress high-resolution (256x256) image patches of the Messier 51 galaxy into an 8-qubit simulation. The theoretical framework established by Heredge is not only valid but highly portable across different quantum programming ecosystems.

IBM Quantum / Qiskit Contributors. (2024). *Qiskit documentation: Quantum machine learning*. Qiskit.org. <https://qiskit.org/ecosystem/machine-learning/>

This source is the official software development kit (SDK) documentation provided by IBM, demonstrating how quantum states can be manipulated programmatically to solve classification and regression problems using quantum support vector machines and variational circuits. As a primary technical resource, it is indispensable for understanding the mechanics beneath the quantum abstraction layer. For the symposium, this documentation acts as the software parallel to understanding classical Big Data frameworks. Just as Hadoop abstracts the complexity of a distributed file system and PySpark abstracts data distribution across classical nodes, frameworks like Qiskit and PennyLane act as critical translators between Python code and complex quantum operations. This source proves that modern quantum hardware relies on these vital abstraction layers to function practically for data scientists executing live algorithms.

Kohler, S. (2018, November 7). Deluges of data are changing astronomical science. *Eos: Earth & Space Science News*.
<https://eos.org/features/deluges-of-data-are-changing-astronomical-science>

Published by the American Geophysical Union, this feature article outlines the immediate logistical nightmare facing modern observatories, arguing that the sheer volume of incoming high-resolution data has entirely outpaced standard analytical capabilities. The article provides a highly readable, accessible overview of the problem without relying on extreme technical jargon. Within the symposium, this source serves as the historical baseline. It explains the exact structural conditions that forced the astronomy sector to initially adopt massive, distributed data engineering frameworks like Google's MapReduce and Hadoop. By clearly outlining this initial data crisis, it perfectly sets up the presentation's argument that these classical models have once again reached their saturation point, necessitating the current transition to Quantum Machine Learning.

Marchetti, L., et al. (2025). Quantum computing for space applications: a selective review and perspectives. *EPJ Quantum Technology*, 12, 15.

<https://doi.org/10.1140/epjqt/s40507-025-00369-8>

This highly current, peer-reviewed article synthesizes recent literature to argue that Quantum Machine Learning (QML) offers distinct computational advantages specifically for Earth observation, scheduling, and processing broad spatial data. Because the challenges in Earth observation map neatly onto astronomical survey pipeline problems, this review validates the transition of standard space-data pipelines into the quantum realm. It strengthens the symposium by highlighting how the technological race for quantum computing is actively shifting how space agencies design their data infrastructure. It proves that the industry is moving away from a strict reliance on classical data clusters and toward hybrid quantum-classical loops to achieve a definitive operational advantage in space science research.

Miller, C. (2022). *Chip war: The fight for the world's most critical technology*. Scribner.

In this comprehensive historical and economic analysis, Chris Miller argues that semiconductors are the fundamental resource defining modern global power, positing that the global supply chain is perilously centralized and highly susceptible to weaponization. This book provides macro-level political framing for the entire symposium. The presentation asserts that one cannot run a massive Google MapReduce pipeline or a PySpark cluster without thousands of classical silicon chips. Miller's analysis supports the presentation's "Material Monopoly" argument, explaining the severe implications of China controlling the vast majority of the world's Gallium and Germanium. By clearly defining the constraints of classical silicon dependency, this text establishes exactly why global superpowers are engaged in a strategic standoff and are desperately rushing to develop post-silicon quantum architectures.

U.S. Congressional Research Service. (2024). *U.S. export controls and China: Advanced semiconductors*. congress.gov/crs-product/R48642

This official government report details the economic and legislative landscape of U.S. export controls targeting advanced semiconductor technology, arguing that maintaining a national technological edge requires strict control over the global supply chain. As an objective Congressional Research Service document, it is a highly authoritative policy source. This report perfectly anchors the "Geopolitical Gatekeeping" section of the symposium. Building the massive classical server farms required to run enterprise-scale Hadoop infrastructure depends entirely on vulnerable manufacturing pipelines. This document outlines real-world manufacturing blockades - such as bans on advanced ASML lithography equipment - illustrating how the politics of classical chips immediately restrict the scaling of traditional data architectures, thereby forcing a sovereign race toward alternative hardware solutions.